

1. Actual vs Predicted Total Sales

What the graph shows:

Each blue dot represents a real sale amount vs. the predicted model output.

The red dashed line represents the ideal “perfect prediction.”

Your points lie very close to the red line, meaning the model predictions match the actual values almost perfectly.

Key Metrics:

- MAE ~ 34
- RMSE ~ 53
- $R^2 = 0.9982$

Interpretation:

The regression model understands your sales behaviour extremely well, even though the dataset is highly spiky.

The model is successfully capturing transaction patterns, time-based behaviour, and sales variability.

Future Outcome:

- Highly accurate sales forecasting
- Better inventory planning
- Anticipation of sales spikes
- Reliable profit estimation

2. High-Value Transactions Distribution

What the graph shows:

The bar chart compares:

0 → Normal transactions

1 → High-value transactions

Most transactions are normal, while high-value ones are fewer.

Interpretation:

Your business runs on high volume of small or medium purchases.

High-value purchases are rare but important, and they contribute major revenue spikes.

Future Outcome:

- Identify future high-value customers
- Enable VIP targeting
- Detect unusual purchasing behaviour
- Plan premium-level offers

3. Daily Total Sales Trend (Highly Spiky Dataset)

What the graph shows:

Blue line = daily sales (very spiky)

Red line = 7-day rolling average (smoothed trend)

Interpretation:

The dataset shows:

- Seasonal peaks
- Hourly/daily spikes
- Stable long-term trend

Future Outcome:

- Expect continued spiky but consistent growth
- Forecast peaks weekly/monthly
- Use forecasting models like XGBoost, LSTM, Prophet for planning

Overall Future Business Insights

1. Sales Forecasting:

- Predict next 7/30 days of sales
- Improve stock planning
- Reduce stockouts & wastage

2. Customer Behavior Prediction:

- Predict future high-value buyers
- Detect upcoming demand surges
- Optimize promotional strategies

3. Operational Optimization:

- Schedule staff during peak hours
- Prepare inventory for high-demand days
- Reduce operational bottlenecks

4. Power BI Dashboard Integration:

- Future sales forecast dashboards
- High-value customer alerts
- Time-of-day heatmaps
- Predictive business KPIs